www.tntextbooks.in

Activity

- Create a model of the ear by using clay, recyclables etc.
- Use different colors to indicate different lobes of the brain.
- Draw the different areas of the cerebral cortex on the white swimming cap with different colour markers.

Summary

Neural system coordinates and integrates the functions of all organs and responds to changes in the internal and external environments.

Neural system includes two types of cells neurons and neuroglia. Neuron forms the structural and functional unit of the neural system.

CNS includes brain and spinal cord. The major parts of the brain are the cerebrum, diencephalon, cerebellum and the brain stem. The brain is protected by the cranium and meninges. CSF provides mechanical protection and nutrients to the CNS.

The spinal cord is the continuation of the medulla oblongata and ends at the second lumbar vertebra as conus medullaris. The components involved in reflex action are called reflex arc.

There are 12 pairs of cranial nerves which arise from the brain and 31 pairs of spinal nerves from the spinal cord make the PNS. The PNS is further divided into Somatic Neural system and Autonomic Neural system. The SNS operates under conscious control. The ANS usually operates unconsciously.

The motor portion of ANS has two major divisions: sympathetic and parasympathetic.

Sensation is the conscious or subconscious awareness of changes in the external or internal environment. Four events namely stimulation, transduction, generation of impulses and integration occurs typically for a sensation to take place.

Simple receptors are associated with the general senses such as somatic senses (skin); complex receptors are associated with the special senses such as smell, taste, vision, hearing and equilibrium.

A person on his way to a village in a car for weekend holidays after finishing the office work. As he is very tired, he begins to feel drowsy. He turns up the car stereo volume, opens the car window and has sips of ice-cold water. How do these actions keep him awake? The increase in the number of sensory stimuli he receives is relayed to the cerebral cortex which gets activated and prevents sleeping.